Lab 9: Cell Division — Mitosis & Meiosis

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

Cells must divide to grow, repair tissues, and reproduce. Eukaryotic cells have two methods of division: **Mitosis** (for identical copies) and **Meiosis** (for genetically unique gametes). In this lab, you will compare these two processes using microscopy and modeling.

Learning Objectives

By the end of this lab, you will be able to:

1. Identify the phases of mitosis (Prophase, Metaphase, Anaphase, Telophase) in plant cells.
2. Model the movement of chromosomes during meiosis.
3. Demonstrate how independent assortment and crossing over create genetic diversity.
4. Contrast the outcomes of mitosis (2 diploid cells) and meiosis (4 haploid cells).

Materials

- Compound Microscope
- Prepared slides of **Onion Root Tip** (*Allium* root tip)
- Chromosome Modeling Kit (Pop beads or Pipe cleaners)
- Colored pencils

Part 1: Observing Mitosis in Onion Root Tips

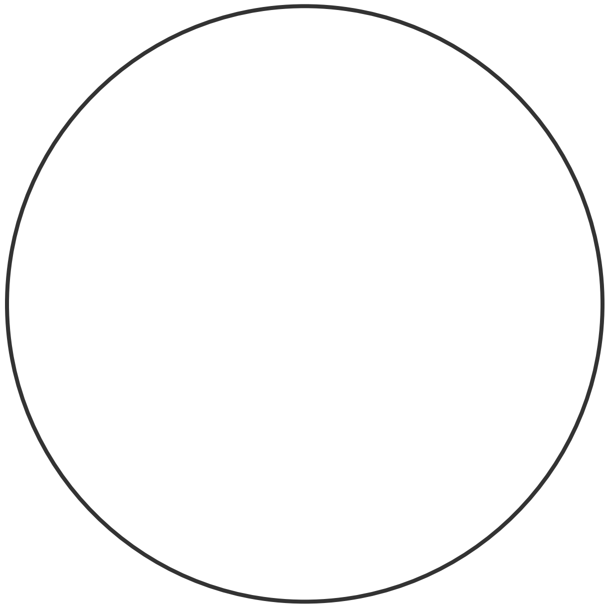
The root tip is a region of rapid growth, so many cells are dividing.

Procedure:

1. Focus on the onion root tip slide under high power (40× objective).
2. Scan the cells just behind the root cap. Look for cells with visible condensed chromosomes.

3. Identify one cell in each phase: **Interphase, Prophase, Metaphase, Anaphase, Telophase**.

Draw one cell in METAPHASE and one cell in ANAPHASE.



Label the chromosomes and the spindle fibers (if visible).

Part 2: Modeling Meiosis

Use the chromosome kit to simulate meiosis.

- **Start:** A diploid cell (2n = 4). Use two long beads (one red, one blue) and two short beads (one red, one blue).
- **Red** = Maternal chromosomes; **Blue** = Paternal chromosomes.

Procedure:

1. **Interphase:** Replicate your DNA. Create sister chromatids for each chromosome.
2. **Prophase I:** Pair up homologous chromosomes (Red Long with Blue Long, Red Short with Blue Short).
    - **Perform Crossing Over:** Swap a few beads between the non-sister chromatids of the long pair.
3. **Metaphase I:** Line up the pairs at the center. (Randomly! Red left/Blue right OR Blue left/Red right).
4. **Anaphase I:** Separate the **homologous pairs**.
5. **Meiosis II:** Separate the **sister chromatids**.

1. At the end of Meiosis II, how many cells do you have? Are they haploid or diploid?

2. Are the daughter cells genetically identical to the parent cell? Why or why not?

3. How did independent assortment (random alignment in Metaphase I) affect the combination of red/blue chromosomes in the final cells?

Conclusion

Mitosis preserves genetic identity for growth, while Meiosis generates diversity for sexual reproduction. The physical movements of chromosomes explain Mendel's laws of inheritance.